

IMPORT LIBRARIES

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

LOAD THE DATASET

```
In [2]: df = pd.read_csv("Admission_Predict.csv")
print(df.head())
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	\
0	1	337	118	4	4.5	4.5	9.65	
1	2	324	107	4	4.0	4.5	8.87	
2	3	316	104	3	3.0	3.5	8.00	
3	4	322	110	3	3.5	2.5	8.67	
4	5	314	103	2	2.0	3.0	8.21	

	Research	Chance of Admit
0	1	0.92
1	1	0.76
2	1	0.72
3	1	0.80
4	0	0.65

DATA PRE-PROCESSING

```
In [3]: df = df.drop(['Serial No.'], axis=1)
df['Admitted'] = df['Chance of Admit '] >= 0.5
df['Admitted'] = df['Admitted'].astype(int)
```

FEATURE SELECTION

```
In [4]: X = df[['GRE Score', 'TOEFL Score', 'CGPA']]
y = df['Admitted']
```

TRAIN-TEST-SPLIT

```
In [5]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

TRAIN DECISION TREE MODEL

```
In [6]: model = DecisionTreeClassifier(max_depth=3)

model.fit(X_train, y_train)
```

Out[6]:

DecisionTreeClassifier ⓘ ?

Parameters

MODEL PREDICTION

```
In [7]: y_pred = model.predict(X_test)
```

MODEL EVALUATION

```
In [8]: print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.925

Confusion Matrix:

```
[[ 4  6]
 [ 0 70]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.40	0.57	10
1	0.92	1.00	0.96	70
accuracy			0.93	80
macro avg	0.96	0.70	0.77	80
weighted avg	0.93	0.93	0.91	80

VISUALIZATION OF DECISION TREE

```
In [9]: from sklearn.tree import plot_tree

plt.figure(figsize=(10,6))
plot_tree(model, feature_names=X.columns, class_names=['No', 'Yes'], filled=True)
plt.show()
```

